Dietary cadmium exposure and risk of postmenopausal breast cancer: a population-based prospective cohort study.

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The ubiquitous food contaminant cadmium has features of an estrogen mimetic that may promote the development of estrogen-dependent malignancies, such as breast cancer. However, no prospective studies of cadmium exposure and breast cancer risk have been reported. We examined the association between dietary cadmium exposure (at baseline, 1987) and the risk of overall and estrogen receptor (ER)-defined (ER(+) or ER(-)) breast cancer within a population-based prospective cohort of 55,987 postmenopausal women. During an average of 12.2 years of follow-up, 2,112 incident cases of invasive breast cancer were ascertained (1,626 ER(+) and 290 ER(-)). After adjusting for confounders, including consumption of whole grains and vegetables (which account for 40% of the dietary exposure, but also contain putative anticarcinogenic phytochemicals), dietary cadmium intake was positively associated with overall breast cancer tumors, comparing the highest tertile with the lowest [rate ratio (RR), 1.21; 95% confidence interval (CI), 1.07-1.36; P(trend) = 0.02]. Among lean and normal weight women, statistically significant associations were observed for all tumors (RR, 1.27; 95% CI, 1.07-1.50) and for ER(+) tumors (RR, 1.25; 95% CI, 1.03-1.52) and similar, but not statistically significant associations were found for ER(-) tumors (RR, 1.22; 95% CI, 0.76-1.93). The risk of breast cancer increased with increasing cadmium exposure similarly within each tertile of whole grain/vegetable consumption and decreased with increasing consumption of whole grain/vegetables within each tertile of cadmium exposure (P(interaction) = 0.73). Overall, these results suggest a role for dietary cadmium in postmenopausal breast cancer development.